

SURGICAL AUDIT & RESEARCH

Clinical audit — definition

- **Quality improvement process** — improve patient care + outcomes through **systematic review of care against explicit criteria + review of change** (NICE 2002).
- "Auditing" from Latin "**audire**" = "**to hear**".
- Quote (Devlin): "surgery without audit is like playing cricket without keeping the score."
- **Pioneer: Florence Nightingale** (Crimean War) — sanitary standards + meticulous records → mortality 40% → 2%.

Care aspects (audit compares against criteria)

- **Structure** — what's in place (people, training, equipment, organization).
- **Process** — what you do (referrals, diagnostic tests, antibiotics, thromboprophylaxis, feeding/mobilization policy, discharge).
- **Outcome** — results (wound dehiscence, readmission, mortality, QOL, return to work).

Explicit criteria

- If care falls short → change proposed at: individual (training), instrument (replace), team (nurses doing procedures), institution (new antibiotic policy), regional (tertiary referral centre), national (screening programmes).

The audit cycle

1. **Define audit question.**
 2. Identify evidence + current standards.
 3. **Design audit** to measure performance vs agreed standards.
 4. Measure over agreed interval (collect data).
 5. **Analyze + compare** vs standards.
 6. **Gap analysis:** standards met → re-audit after interval; need improvement → intervene (training) + agree with parties.
 7. **Re-audit.**
- (Loop: problem identified → criteria/standards set → data collected → identify improvement areas → make changes → re-audit.)

Research

- **Definition:** systematic investigation to discover facts/relationships + reach conclusions using **scientifically sound methods**.
- **Clinical audit is NOT research.**
- **Study types:** observational (retrospective/prospective), **case-control** (disease vs matched controls), cross-sectional (single occasion), longitudinal (over time), **experimental** (compare treatments, allocation controlled), **randomized, randomized controlled** (+ control group).
- **Research identification:** purpose = new knowledge/set standards; methods = RCTs; extensive statistics; **always needs ethics + trust approval**; powered sample size; hypothesis-driven; generalizable + publishable.

Research vs Audit (key distinction)

- **Research asks:** "Are we singing the **RIGHT** song?" (sets standards / new knowledge).
- **Audit asks:** "Are we singing this song **RIGHT**?" (measures against existing standards).

High-yield one-liners

- **Audit = systematic review against explicit criteria (structure/process/outcome); NICE 2002; Nightingale pioneer.**
- **Audit cycle: question → standards → measure → compare → gap analysis → change → re-audit.**
- **Research = new knowledge (needs ethics approval); audit = measure against existing standards (no ethics approval).**
- **Research = "right song?"; Audit = "song right?"**

SURGICAL AUDIT & RESEARCH — Revision Table

Bold = high-yield. Prof. Kamal Saeed.

CLINICAL AUDIT

Topic	Key points
Definition	Quality improvement — systematic review against explicit criteria (NICE 2002); "audire" = to hear
Pioneer	Florence Nightingale (Crimea) → mortality 40% → 2%
Care aspects	Structure (what's in place), Process (what you do), Outcome (results)
Change levels	Individual, instrument, team, institution, regional, national
Audit cycle	Question → standards → measure → compare → gap analysis → change → RE-AUDIT

RESEARCH

Topic	Key points
Definition	Systematic investigation for new facts using scientifically sound methods
Audit ≠ research	Audit is NOT research
Study types	Observational (retro/prospective), case-control, cross-sectional, longitudinal, RCT
Identification	New knowledge; RCTs; always needs ethics/trust approval ; powered sample; hypothesis-driven; publishable

RESEARCH vs AUDIT

	Research	Audit
Question	"Are we singing the RIGHT song?"	"Are we singing this song RIGHT?"
Purpose	New knowledge / set standards	Measure vs existing standards
Ethics approval	Always required	Not required